

Answer Part 1

Git Use with Engineering Workflow

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Author feature-branch review snapshot. Final hardening approval, reviewed staging-to-main integration, final-main output, final Project Done evidence and student model/Reflection confirmation are pending. This PDF is not the final submission.

[Repository](#) | [Project board](#) | [Six work items](#) | [Remaining release gates](#)

Rendered repository source: docs/lab-04/reviewer.md

Lab 4 Review Record

Status: contract #39, data foundation #41, Actions #43, workflow #45 and dashboards #47 are independently approved and merged into `codex/lab4-staging`; Issues #38/40/42/44/46 are closed/Done. Datakung approved exact dashboard correction head `884df39` on 2026-10-10 at 03:41:12 Asia/Bangkok and merged at 03:41:27 (`72c5553`). The re-review confirmed both normal test/capture commands pass without the Prisma client-path workaround and no blocking findings remain. Final hardening/release #48 is in progress; author checks do not count as peer approval.

Engineering contract issue: [#38](#). Local document checks and author review do not count as independent peer approval.

Selected current record; historical findings and correction replies are retained in the linked full source.

Area	Reviewer	PR	Comment/response	Approval
Engineering contract	Datakung	#39	Changes requested, four corrections, author reply and re-review; no remaining blocking findings	Approved <code>4886585</code> ; merged by Datakung
Data foundation / recovery	Datakung	#41	Reported independently passing 391 tests and both builds; no blocking findings	Approved <code>301c902</code> ; merged
Actions API/UI	Datakung	#43	Independently passed 431 tests, seven browser journeys and both builds at <code>935bcdf</code> ; all findings corrected	Approved <code>935bcdf</code> ; merged <code>68a88cc</code>
Ticket workflow	Datakung	#45	Independently confirmed 458 tests, 14 browser cases and both builds; no blocking findings	Approved <code>a1d1fce</code> ; merged <code>69b32d6</code>
Dashboards and UI	Datakung	#47	Changes requested on <code>1f1adc2</code> : E2E fixture resolved an ungenerated root Prisma client. Fixed server client resolution; author response records both commands passing four cases each without workaround. Peer re-reviewed and found no blockers	Approved <code>884df39</code> ; merged <code>72c5553</code>
Final regression	Pending	Pen ding	Pending	Pending

Actual commit history before final release

- * `5172f83` fix(lab4): keep follow-up checkbox compact and match Zen Green theme
- * `4ab9c79` test(lab4): capture Action forms and dashboard feedback for final review
- * `403e044` test(lab4): scope polish error checks to authenticated routes
- * `7ec3be3` test(lab4): add final regression and dashboard scale verification
- * `72c5553` Merge pull request #47 from auto4496/codex/lab4-5-role-dashboards
- * `69b32d6` Merge pull request #45 from auto4496/codex/lab4-4-ticket-workflow
- * `68a88cc` Merge pull request #43 from auto4496/codex/lab4-3-actions

```
* c039a7d Merge pull request #41 from auto4496/codex/lab4-2-data-foundation
* b5bee16 Merge pull request #39 from auto4496/codex/lab4-1-engineering-contract
* da5a566 Merge pull request #37 from auto4496/codex/lab3-copy-corrections
```

README, ignore rules and directory structure

[Current README](#) documents LAB4 setup, data-preserving upgrade, guarded tests, all-suite capture and the six-step demo. [.gitignore](#) excludes credentials, dependencies, builds, temporary files, generated test-results and local report output. Curated evidence is explicitly published after successful verification.

```
client/src/{Dashboard,ActionsTaken,StaffTicketDetail}.tsx
client/tests/lab-04/
server/src/{actions-taken,dashboards,workflow-rules}.ts
server/prisma/{schema.prisma,migrations/}
server/tests/lab-04/
e2e/lab-04/
docs/lab-04/
artifacts/lab-04/{actions,workflow,dashboards,release}/
scripts/lab-04/build-review-report.cjs
```

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Answer Part 2

Spec DD

The engineering contract existed in commit dd34049 and was independently approved through [PR #39](#) before foundation and feature completion. Later implementation/review corrections remain identifiable in Git history.

Rendered repository source: [docs/lab-04/specification.md](#)

Lab 4 Engineering Contract

Status: contract [Issue #38](#) approved by Datakung at 4886585 and integrated through [PR #39](#). The initial contract was committed separately in dd34049 before feature implementation. Baseline: integrated Lab 3 origin/main at da5a566. Data foundation #40/#41, Actions #42/#43, workflow #44/#45 and dashboards #46/#47 are approved/merged. Final hardening and release [#48](#) starts from reviewed dashboard merge 72c5553.

1. Sprint goal

Complete the service desk lifecycle with accountable Actions Taken, a server-enforced resolution gate, and concise role dashboards while preserving Lab 1-3 behavior.

2. Stakeholder request

Staff plan and record individual work under a Ticket. The Ticket Owner coordinates the case, while another eligible staff member can be assigned to an Action and perform it. Requesters can see Actions and indicate apparent resolution; staff formally change Ticket status. Dashboards lead to the existing detailed screens.

3. Scope

Included: Action model, immutable revision history, create/assign/edit/complete/cancel/list, final Ticket status enforcement, Requester and Staff dashboards, additive migration, idempotent demo seed, responsive Zen Green UI, authorization and regression. Admins may perform staff Action operations and use the Staff dashboard. Lab 3 session and role rules remain authoritative.

Excluded: SLA/escalation, notifications, inventory, billing, approvals, BI/export, multi-tenancy and production cloud operations. Attachment Notes refer to existing Ticket files; Actions add no separate file upload.

4. Functional requirements

ID	Requirement
FR-01	Staff/Admin list Actions on any Ticket; Requesters read Actions only on owned Tickets.
FR-02	Staff/Admin create an Action with date/time, description, result, assignee, follow-up flag/note and attachment notes; server records actor.
FR-03	Staff/Admin edit, reassign, complete or cancel a nonterminal Action with a version check.
FR-04	Every Action mutation appends an immutable revision with actor, time, operation and resulting values.
FR-05	Formal Ticket resolution requires a valid Lab 3 transition and the Action completion gate.
FR-06	Requester dashboard shows owned open, waiting, recently updated and recently resolved Tickets.
FR-07	Staff dashboard shows unassigned, owned, status/IT Priority counts, follow-up Actions and recent Tickets. Admin may use it.
FR-08	Cards and recent entries lead to Ticket list/detail, with loading, zero, forbidden, conflict and failure feedback.
FR-09	Existing authentication, user management, conversations, attachments and Ticket journeys remain functional.

5. Business rules

ID	Rule
BR-01	One Action belongs to exactly one Ticket. Restrictive foreign keys preserve history.
BR-02	Ticket Owner, Action assignee and performer may be different eligible staff members.
BR-03	Server sets creator, latest performer and audit times from the authenticated session/clock; client overrides are rejected.
BR-04	Action date/time is a required ISO 8601 instant selected by staff, stored/returned in UTC; past and future are allowed.
BR-05	Description is 1-2000 Unicode code points. Result is 0-2000 until completion and 1-2000 on completion. Optional notes are at most 2000; normalize NFC, CRLF to LF, and trim.
BR-06	Follow-up Note is required when Follow-Up Required is true; clear it when false.
BR-07	A new assignee must be active IT_STAFF or ADMINISTRATOR. Historical inactive assignees remain visible.
BR-08	Action statuses: PLANNED → IN_PROGRESS/COMPLETED/CANCELLED; IN_PROGRESS → COMPLETED/CANCELLED. Completed/cancelled rows are immutable.
BR-09	Staff/Admin can work on any Ticket; CLOSED/CANCELLED Tickets reject Action writes. Requesters never write Actions.
BR-10	Edit/transition requires expectedVersion; transaction row lock and version comparison return 409 on stale writes. Creation uses Idempotency-Key to make retries safe.
BR-11	Successful Action mutations append a revision atomically; no revision edit/delete API exists.

ID	Rule
BR-12	Only IT_STAFF changes Ticket status. Admin Action permissions do not grant Ticket status permission. Existing owner/confirmation/version conditions remain.
BR-13	RESOLVED additionally requires at least one COMPLETED Action with nonblank Result and zero PLANNED/IN_PROGRESS Actions. Requester indication stays advisory.
BR-14	CLOSED requires RESOLVED; reopening retains Actions. CANCELLED is terminal.
BR-15	Dashboard open excludes RESOLVED/CLOSED/CANCELLED. Recent means updatedAt >= now - 7 days using a UTC instant from one server clock snapshot.
BR-16	Recent lists contain at most five rows sorted updatedAt DESC, id DESC; empty counts are zero and lists are [].
BR-17	Dashboard links use Ticket list filters, including the documented open-status and Action-assignee scopes; responses expose no Internal Notes or whole Ticket collections.
BR-18	Existing Tickets need no backfill: zero Actions is valid history, but they cannot newly enter RESOLVED. Existing RESOLVED/CLOSED Tickets remain as recorded.

Final Ticket transition matrix

From	Permitted next statuses
NEW	OPEN, CANCELLED
OPEN	IN_PROGRESS, WAITING_FOR_REQUESTER, CANCELLED
IN_PROGRESS	WAITING_FOR_REQUESTER, RESOLVED, CANCELLED
WAITING_FOR_REQUESTER	IN_PROGRESS, RESOLVED, CANCELLED
RESOLVED	CLOSED, REOPENED
CLOSED	REOPENED
REOPENED	OPEN, IN_PROGRESS, WAITING_FOR_REQUESTER, CANCELLED
CANCELLED	none

6. UI specification summary

[ui-spec.md](#) defines Dashboard and Actions layout, form modes, role behavior, mobile arrangement, keyboard/focus behavior and feedback. Status and priority use text as well as color. Requesters see read-only Actions.

7. Data changes

Add `ActionTaken` (UUID, Ticket/assignee/creator/latest-performer FKs, action time, normalized text, follow-up fields, status, version and timestamps) and `ActionRevision` (Action/actor FKs, operation, immutable value snapshot and timestamp). Index (ticketId, actionAt, id), (ticketId, status), (assigneeId, status). An integer version follows existing Ticket concurrency behavior. A separate revision table keeps the current Action compact and retains append-only evidence of edits; these are the two main design decisions.

Migration is additive and does not delete/backfill Ticket rows. Before migration, back up the database and attachment directory and record counts. Apply first to isolated test DB; verify User, Ticket, Attachment, Comment and Note counts and FKs. On failure stop writes and restore the pre-migration database backup and attachment copy into a guarded clean recovery target; compare records, relationships and attachment checksums before accepting recovery. INT-02 in [tests.md](#) defines the controlled isolated failure/restore scenario and required evidence. Seed deterministic examples across statuses, priorities, assigned/unassigned Tickets and zero/one/multiple Actions; insert only missing rows, never overwrite edits/passwords.

8. API contract

[api-spec.md](#) gives exact paths, fields, response shapes, role/ownership checks, conflicts, ordering and calculations. All writes retain Lab 3 cookie session and CSRF enforcement.

9. Acceptance criteria

ID	Criterion
AC-01	Staff/Admin create an Action under the right Ticket with server actor and active assignee.
AC-02	Requester reads owned Actions, cannot read another Ticket or mutate Actions.
AC-03	Invalid fields/inactive assignee leave no Action or revision.
AC-04	Edit/transition records ordered revisions; stale/terminal updates fail safely.
AC-05	Same-key retries create one Action; same key with different payload conflicts.
AC-06	Every Ticket status edge and the resolution gate are enforced by the backend.
AC-07	Requester dashboard matches owned database queries without leakage.
AC-08	Staff dashboard counts/order match queries, and drill-down reaches matching Tickets.
AC-09	Migration preserves Lab 3 data and seed reruns do not overwrite records.
AC-10	Desktop/tablet/mobile UI handles loading, empty, invalid, conflict, forbidden and failure states accessibly.
AC-11	Full Lab 1-3 authentication, ownership, attachment, conversation, staff and admin regression passes.

10. Product Definition of Done

Every AC maps to passing tests in [tests.md](#); backend role, ownership and concurrency checks pass; migration/seed are verified on the isolated test DB; full build/regression and desktop/tablet/mobile/keyboard checks pass; README/demo/evidence are current; peer review is resolved; final submission claims reflect observed results.

11. Assumptions and decisions

The handout does not specify Action statuses, assignee representation, dashboard window or precise resolution gate. This contract chooses explicit statuses, one current assignee, seven days UTC and one completed Action with Result before resolution. Ticket status authority remains Lab 3's. The contract and all five feature work items are independently approved and integrated into staging; final hardening/release remains #48.

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Answer Part 3

Test DD and Traceability

All results below are author feature-branch results. Final-main output must be added after the actual reviewed main merge.

484 tests / 48 files; 32 complete browser cases; both builds passed. Source commits and command history: [observed verification](#).

Lab 4 Test Plan and Traceability

Status: contract/data/Actions/workflow/dashboards independently approved and merged; Issue #48 has final author verification with 48 files / 484 passing tests, 32 complete browser cases and both builds. Independent hardening review and final-main release remain pending. Pending means no passing result has been observed in the staged work items. Use only isolated TEST_DATABASE_URL and the Lab 3 guard; never run destructive test setup against the development database.

The matrix records actual implemented checks; full author execution and exact provenance are in [final-verification.md](#). See [data foundation evidence](#), [Actions evidence](#), [workflow evidence](#) and [dashboard evidence](#). Pending filenames remain proposed implementation locations. Author execution does not imply peer approval or staged acceptance.

ID	Type	AC	Check	Expected	Automated file	Final
UNIT-01	API coverage (planned unit)	AC-03	Normalize Action text, follow-up and completion fields	Invalid/over-limit rejected	server/tests/lab-04/actions-taken.api.test.ts; no separate unit file claimed	Pass through API
UNIT-02	Unit	AC-06	Complete transition matrix, confirmation/owner/advisory sets and Action gate decision	Exact matrix/gate	server/tests/lab-04/ticket-workflow.unit.test.ts	Pass: seven checks
API-01	API	AC-01, AC-03	Create Action, actor/assignee, validation and inactive assignee	Correct data/400	server/tests/lab-04/actions-taken.api.test.ts	Pass
API-02	API	AC-02	Own/non-owned Requester reads and write denial	Own only/403/404	server/tests/lab-04/actions-taken.api.test.ts	Pass
API-03	API	AC-04	Edit, transition, revisions, stable list order, terminal lock	Ordered history/409	server/tests/lab-04/actions-taken.api.test.ts	Pass
API-04	API	AC-04, AC-05	Concurrent stale edit and repeated create key	One winner/one row	server/tests/lab-04/actions-taken.api.test.ts	Pass
API-05	API	AC-06	Status matrix, owner/confirmation/version and resolution gate via raw request; committed/rolled-back competing Action transaction	Only legal transitions; no partial Ticket update	server/tests/lab-04/ticket-workflow.api.test.ts plus retained server/tests/lab-03/staff-workflow.api.test.ts (all 64 edges)	Pass: new gate cases plus retained matrix
API-06	API	AC-07	Requester counts/window/order/other-user isolation; live concurrent write during repeatable-read snapshot	Database-matching owned data	server/tests/lab-04/requester-dashboard.api.test.ts	Pass: five checks
API-07	API	AC-08	Staff/Admin metrics, role denial, deterministic recent list, intersecting open/follow-up filters and safe failure	Exact counts/order/403	server/tests/lab-04/staff-dashboard.api.test.ts	Pass: eleven checks
INT-01	Integration	AC-09	Migrate Lab 3 fixture and compare counts/FKs; seed twice	No loss/overwrite	server/tests/lab-03/migration.integration.test.ts (retained suite extended through Lab 4)	Pass
INT-02	Integration	AC-09	Back up a Lab 3 fixture and attachment files; trigger controlled migration/validation failure; restore into an isolated recovery target	Baseline records, relationships and attachment checksums preserved	server/tests/lab-04/migration-recovery.integration.test.ts	Pass

ID	Type	AC	Check	Expected	Automated file	Final
UI-01	Component	AC-10	Staff/Admin dashboard cards, links, zero/error and actual list filter state	Accessible states	client/tests/lab-04/StaffDashboard.test.tsx; DashboardRouting.test.tsx	Pass
UI-02	Component	AC-10	Requester dashboard ownership display, links, loading/error/forbidden and filtered authentication destination	Accessible states	client/tests/lab-04/RequesterDashboard.test.tsx; DashboardRouting.test.tsx	Pass
UI-03	Component	AC-01-04, AC-10	Action read/edit/create, validation, conflict, role visibility	Safe UI states	client/tests/lab-04/ActionsTaken.test.tsx	Pass
UI-04	Component	AC-06, AC-10	Readiness/role gating, busy protection, gate error/focus, Action refresh, failed refresh and ambiguous status recovery	Correct feedback and retained selections	client/tests/lab-04/TicketWorkflow.test.tsx	Pass: nine checks
UI-05	Component	AC-04, AC-10	Start visibility by role, Action/Ticket status; pending click protection, success refresh, safe failure and stale-version reload	PLANNED becomes IN_PROGRESS once; Requester/terminal writes unavailable	client/tests/lab-04/ActionsTaken.test.tsx	Pass
STYLE-01	Style/browser /component	AC-10	Focus, labels, text status, mobile card wrapping	No inaccessible controls	Actions/workflow/dashboard component/browser checks and screenshots; proposed standalone style file absent	Pass in author scope: full retained style/browser checks, new ten-route focus/error check and three-width form validation; see visual/accessibility checklist . Student zoom/final-main remain release gates
E2E-01	E2E	AC-01-05	Two staff, one Ticket, multiple Actions, revisions	End-to-end work	e2e/lab-04/actions-taken-flow.spec.ts	Pass: two IT Staff plus Admin; final focused rerun
E2E-02	E2E	AC-06	Advisory indication, Staff resolution, close/reopen preservation and real competing unfinished Action; terminal cancel checks retained in Actions suite	Gate enforced with usable retry	e2e/lab-04/ticket-resolution.spec.ts plus retained action-start.spec.ts	Pass: two workflow journeys and seven retained Actions checks
E2E-03	E2E	AC-07-08, AC-10	Requester/Staff/Admin, real filtered list queries/detail, Back/Forward, keyboard activation, loading/retry/empty/forbidden and three widths	Correct responsive views	e2e/lab-04/dashboards.spec.ts	Pass: four journeys, nine published screenshots
E2E-04	E2E	AC-04, AC-10	Staff and Admin use Start; verify IN_PROGRESS, performer and STARTED revision; Requester reads without Start; terminal denial and competing-version conflict	Correct lifecycle and recoverable UI feedback	e2e/lab-04/action-start.spec.ts	Pass

ID	Type	AC	Check	Expected	Automated file	Final
E2E-05	E2E	AC-04-05, AC-10	Lose committed or validation-rejected create response, edit draft, recover original body/key; edit recovered ID or release rejected draft for fresh creation	Committed replay: one Action with CREATED/EDITED; rejected retry: one Action with CREATED only, Save/Cancel usable	e2e/lab-04/action-create-recovery.spec.ts	Pass: both real-browser recovery paths
REG-01	Regression	AC-11	All Lab 1-3 Vitest, Playwright, build	Pass with isolated DB	Existing suites and npm run build:*	Author pass: 48 files / 484 Vitest tests, full 32-case integrated browser suite (including all retained Lab 2/3 visual cases) and both builds. Actual final-main run remains a release gate
PERF-01	Smoke	AC-08	Seed-scale dashboard and indexed query plan	No unbounded Ticket payload/query	server/tests/lab-04/dashboard-smoke.integration.test.ts	Pass: real 5,000 Tickets / 10,000 Actions, exact metrics, bounded query/DTO and six natural EXPLAIN plans

Every AC appears in the matrix. Author results are observed; independent final review and final-main acceptance are tracked separately in [release.md](#). The new `e2e/lab-04/release-polish.spec.ts` adds real-route console/API/focus checks and required-field/draft-preservation/form evidence. Manual visual review at 1440×900, 834×1112, and 390×844 records dashboard and Action screenshots plus keyboard/focus findings; it supplements, rather than replaces, the automated checks.

Complete unit/API/integration/component output

```
> toktickit-monorepo@1.0.0 test
> vitest run

RUN v3.2.7 C:/Users/User/toktickit/tmp/lab4-final

✓ server/tests/lab-03/migration.integration.test.ts (4 tests) 15024ms
  ✓ Lab 2 through Lab 4 data migrations > preserves historical identities and attachment/idempotency relations and initializes passwords resumably 4610ms
  ✓ Lab 2 through Lab 4 data migrations > stops on normalized-email collisions without renaming or merging accounts 2803ms
  ✓ Lab 2 through Lab 4 data migrations > seeds all roles and varied tickets, preserving edits, inactive accounts and changed passwords on repeat 7609ms
✓ server/tests/lab-04/migration-recovery.integration.test.ts (1 test) 7268ms
  ✓ INT-02 restores the complete Lab 3 baseline and attachment downloads after controlled Lab 4 validation failure 7266ms
✓ server/tests/lab-04/dashboard-smoke.integration.test.ts (1 test) 6782ms
  ✓ keeps both dashboards bounded and accurate at 5,000 Tickets / 10,000 Actions, with natural query plans 6781ms
✓ server/tests/lab-03/staff-workflow.api.test.ts (127 tests) 4420ms
✓ server/tests/lab-03/auth.api.test.ts (7 tests) 3514ms
  ✓ Authentication API (API-01..04) > gates initial-password sessions, changes password, rotates session and logs out 758ms
  ✓ Authentication API (API-01..04) > does not disclose unknown, inactive or wrong-password accounts 523ms
  ✓ Authentication API (API-01..04) > invalidates expired sessions and sessions whose user version or activity changes 776ms
  ✓ Authentication API (API-01..04) > limits failed login attempts and supplies a retry interval 1154ms
✓ server/tests/lab-03/admin-users.api.test.ts (25 tests) 3179ms
  ✓ API-12-14 administrator accounts > normalizes new users, stores a real hash and handles simultaneous duplicate emails 553ms
  ✓ API-12-14 administrator accounts > resets a password, invalidates every session, requires a different password and enforces the initial gate 1432ms
✓ server/tests/lab-04/actions-taken.api.test.ts (17 tests) 1938ms
✓ server/tests/lab-03/auth.unit.test.ts (8 tests) 1528ms
  ✓ Lab 3 password rules (UNIT-01 / AC-03) > uses independent salts and real scrypt, without trimming passwords or accepting damaged hashes 1085ms
  ✓ Lab 3 password rules (UNIT-01 / AC-03) > does not normalize distinct Unicode passwords into the same password 437ms
✓ server/tests/lab-04/ticket-workflow.api.test.ts (11 tests) 862ms
✓ server/tests/lab-02/attachments.api.test.ts (9 tests) 876ms
✓ server/tests/lab-02/my-tickets.api.test.ts (19 tests) 568ms
✓ server/tests/lab-04/staff-dashboard.api.test.ts (11 tests) 541ms
✓ server/tests/lab-02/create-ticket.api.test.ts (14 tests) 443ms
✓ server/tests/lab-03/authorization.api.test.ts (8 tests) 344ms
✓ server/tests/lab-04/requester-dashboard.api.test.ts (5 tests) 437ms
✓ server/tests/lab-02/ticket-detail.api.test.ts (2 tests) 282ms
✓ server/tests/lab-02/reference-data.integration.test.ts (3 tests) 270ms
✓ server/tests/lab-01/categories.test.ts (1 test) 176ms
✓ server/tests/lab-03/safe-errors.api.test.ts (6 tests) 184ms
✓ server/tests/lab-02/reference-data.api.test.ts (5 tests) 41ms
✓ server/tests/lab-02/requester-context.api.test.ts (6 tests) 34ms
✓ server/tests/lab-01/server.test.ts (1 test) 17ms
✓ server/tests/lab-01/health.test.ts (1 test) 26ms
✓ server/tests/lab-02/ticket-query.unit.test.ts (16 tests) 10ms
✓ server/tests/lab-02/attachment-validation.unit.test.ts (17 tests) 5ms
✓ server/tests/lab-02/test-database-guard.test.ts (6 tests) 4ms
✓ server/tests/lab-02/ticket-validation.unit.test.ts (14 tests) 3ms
✓ server/tests/lab-02/ticket-number.unit.test.ts (3 tests) 4ms
✓ server/tests/lab-02/ticket-idempotency.unit.test.ts (3 tests) 3ms
✓ client/tests/lab-02/ResponsiveStyles.test.tsx (6 tests) 3ms
✓ server/tests/lab-04/ticket-workflow.unit.test.ts (7 tests) 3ms
✓ client/tests/lab-02/ZenGreenStyles.test.tsx (5 tests) 3ms
✓ client/tests/lab-04/ActionsTaken.test.tsx (23 tests) 1135ms
✓ client/tests/lab-03/StaffWorkflow.test.tsx (11 tests) 558ms
```

```

✓ client/tests/lab-03/AdminNavigation.test.tsx (5 tests) 753ms
✓ client/tests/lab-03/UserManagement.test.tsx (6 tests) 575ms
✓ client/tests/lab-04/DashboardRouting.test.tsx (4 tests) 547ms
✓ client/tests/lab-04/TicketWorkflow.test.tsx (9 tests) 583ms
✓ client/tests/lab-02/CreateTicket.test.tsx (8 tests) 593ms
✓ client/tests/lab-02/RequesterTicketDetail.test.tsx (11 tests) 525ms
✓ client/tests/lab-02/MyTickets.test.tsx (6 tests) 504ms
✓ client/tests/lab-03/ConversationRecovery.test.tsx (3 tests) 427ms
✓ client/tests/lab-03/ReviewRegressions.test.tsx (9 tests) 368ms
✓ client/tests/lab-03/ApplicationShell.test.tsx (11 tests) 322ms
✓ client/tests/lab-03/Login.test.tsx (3 tests) 237ms
✓ client/tests/lab-04/StaffDashboard.test.tsx (2 tests) 186ms
✓ client/tests/lab-04/RequesterDashboard.test.tsx (3 tests) 205ms
✓ client/tests/lab-03/ChangePassword.test.tsx (1 test) 130ms

Test Files 48 passed (48)
Tests 484 passed (484)
Start at 04:08:30
Duration 91.34s (transform 893ms, setup 0ms, collect 10.01s, tests 56.44s, environment 13.90s, prepare 4.05s)

```

Complete integrated browser output

```

> toktickit-monorepo@1.0.0 test:e2e:capture:release
> node e2e/lab-04/capture-release-evidence.mjs

[WebServer] (node:17712) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (Use `node --trace-warnings ...` to show where the warning was created)
[WebServer] (node:46428) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (Use `node --trace-warnings ...` to show where the warning was created)
[WebServer] (node:2980) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (Use `node --trace-warnings ...` to show where the warning was created)
Prisma schema loaded from server\prisma\schema.prisma
Datasource "db": PostgreSQL database "lab3_test", schema "lab4_test" at "127.0.0.1:55433"

4 migrations found in prisma/migrations

No pending migrations to apply.

Running 32 tests using 1 worker

(node:43320) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
(Use `node --trace-warnings ...` to show where the warning was created)
ok 1 e2e\lab-02\authentication.spec.ts:5:5 > AUTH-E2E initial-password gate, change, role landing and logout on a
real browser (3.8s)
ok 2 e2e\lab-02\requester-ticket-flow.spec.ts:14:5 > E2E-01 creates a Ticket, finds it in My Tickets, and opens
owned detail (1.5s)
ok 3 e2e\lab-02\requester-ticket-flow.spec.ts:44:5 > E2E-02 switches requester and protects direct Ticket and
Attachment access (2.3s)
ok 4 e2e\lab-02\requester-ticket-flow.spec.ts:78:5 > E2E-03 exercises invalid, limit, download, and soft-removal
Attachment states (2.0s)
ok 5 e2e\lab-02\responsive-visual.spec.ts:25:7 > RESP-01 keeps every requester screen usable at desktop (1.7s)
ok 6 e2e\lab-02\responsive-visual.spec.ts:25:7 > RESP-01 keeps every requester screen usable at tablet (1.8s)
ok 7 e2e\lab-02\responsive-visual.spec.ts:25:7 > RESP-01 keeps every requester screen usable at mobile (1.6s)
ok 8 e2e\lab-02\responsive-visual.spec.ts:66:5 > VIS-01 captures the complete Zen Green screenshot contract (6.4s)
ok 9 e2e\lab-03\admin-users.spec.ts:8:5 > E2E-03 Admin creates, edits and resets an account; new credentials
require password change (4.0s)
ok 10 e2e\lab-03\admin-users.spec.ts:26:5 > RESP/VIS administrator directory, editor and reset dialog at four
widths (3.5s)
ok 11 e2e\lab-03\admin-users.spec.ts:43:5 > Admin dirty edits survive cancelled shell navigation and browser Back
(1.3s)
ok 12 e2e\lab-03\staff-ticket-flow.spec.ts:12:5 > E2E-02 staff workflow, public/private conversation, indication,
Admin restriction and reopening (5.4s)
ok 13 e2e\lab-03\staff-ticket-flow.spec.ts:47:5 > RESP/VIS staff queue and private detail remain usable at desktop,
tablet, mobile and 320px (3.6s)
ok 14 e2e\lab-03\system-verification.spec.ts:15:7 > SYSTEM responsive authentication, recovery and role evidence:
desktop (10.6s)
ok 15 e2e\lab-03\system-verification.spec.ts:15:7 > SYSTEM responsive authentication, recovery and role evidence:
tablet (8.4s)
ok 16 e2e\lab-03\system-verification.spec.ts:15:7 > SYSTEM responsive authentication, recovery and role evidence:

```

```

mobile (7.9s)
  ok 17 e2e\lab-03\system-verification.spec.ts:15:7 › SYSTEM responsive authentication, recovery and role evidence:
narrow (8.0s)
  ok 18 e2e\lab-04\action-create-recovery.spec.ts:6:5 › recovers committed creation after a lost response, retaining
edits without duplicate Actions (1.5s)
  ok 19 e2e\lab-04\action-create-recovery.spec.ts:46:5 › releases a definitively rejected creation after its response
is lost so the corrected draft can be saved (1.5s)
  ok 20 e2e\lab-04\action-start.spec.ts:36:7 › E2E-04 e2e.staff@example.test starts with keyboard/busy protection;
Requester remains read-only (2.8s)
  ok 21 e2e\lab-04\action-start.spec.ts:36:7 › E2E-04 e2e.admin@example.test starts with keyboard/busy protection;
Requester remains read-only (2.1s)
  ok 22 e2e\lab-04\action-start.spec.ts:69:5 › E2E-04 reloads a real competing edit before retrying Start with the
latest version (2.3s)
  ok 23 e2e\lab-04\action-start.spec.ts:94:5 › E2E-04 reloads after an ambiguous network failure and keeps terminal
Actions/Tickets read-only (2.2s)
  ok 24 e2e\lab-04\actions-taken-flow.spec.ts:6:5 › E2E-L4-01 different staff and Admin record Actions visible read-
only to Requester (4.9s)
  ok 25 e2e\lab-04\dashboards.spec.ts:63:5 › E2E-03 Requester Dashboard metrics, filtered lists, history, detail and
responsive keyboard links (1.7s)
  ok 26 e2e\lab-04\dashboards.spec.ts:84:5 › E2E-03 Staff counts and distinct follow-up drill-down preserve
intersecting filters (1.8s)
  ok 27 e2e\lab-04\dashboards.spec.ts:100:5 › E2E-03 Admin Dashboard retains Users and scopes follow-ups to the
current Admin (1.2s)
  ok 28 e2e\lab-04\dashboards.spec.ts:107:5 › E2E-03 Dashboard loading, safe failure, retry, zero and forbidden
states (2.4s)
  ok 29 e2e\lab-04\release-polish.spec.ts:6:5 › RELEASE real role routes have no browser errors, broken API reads or
hidden keyboard focus (4.5s)
  ok 30 e2e\lab-04\release-polish.spec.ts:42:5 › RELEASE Action create, validation and edit forms retain accessible
data at three widths (13.0s)
  ok 31 e2e\lab-04\ticket-resolution.spec.ts:31:5 › E2E-02 indication is advisory; gate refresh enables resolution
then close/reopen preserves Actions (5.5s)
  ok 32 e2e\lab-04\ticket-resolution.spec.ts:61:5 › E2E-02 server rejects stale readiness after a competing Action;
explicit retry refreshes the gate (2.9s)

  32 passed (2.2m)
Published 75 integrated release screenshots with SHA-256 manifest.

```

Production build output

```

> toktickit-monorepo@1.0.0 build:server
> npm --prefix server run build

> toktickit-server@1.0.0 build
> tsc

> toktickit-monorepo@1.0.0 build:client
> npm --prefix client run build

> toktickit-client@1.0.0 build
> tsc && vite build

vite v6.4.3 building for production...
transforming...
✓ 49 modules transformed.
rendering chunks...
computing gzip size...
dist/index.html                                0.47 kB | gzip: 0.31 kB
dist/assets/bootstrap-icons-mSm7cUeB.woff2    134.04 kB
dist/assets/bootstrap-icons-BeopsB42.woff      180.29 kB
dist/assets/index-CHS4LmFP.css                 343.00 kB | gzip: 52.20 kB
dist/assets/index-BOHot_ib.js                  252.72 kB | gzip: 75.15 kB
✓ built in 863ms

```

Scale and recovery evidence

5,000 Tickets / 10,000 Actions: Requester 37.94ms / 1,312 bytes; Staff 30.64ms / 1,568 bytes, five recent rows each. Metrics match independent DB queries; SELECT counts stay five/six. These are single local smoke observations, not production performance guarantees. [Six natural EXPLAIN ANALYZE/BUFFERS plans and counts](#). Full aggregates may correctly scan the fixture.

The full suite repeats real pg_dump/pg_restore recovery, complete record/constraint preservation, attachment hashes, authenticated download and removed-file 404. [Current credential-free recovery evidence](#).

TOKTICKIT / LAB 04 / REVIEW DRAFT

Answer Part 4

AI Use with Reflection

Rendered repository source: <docs/lab-04/ai-use.md>

Lab 4 AI Use

Tool: Codex desktop with OpenAI models. The user changed the model during LAB4. Exact before/after model names are awaiting the student's confirmation; do not infer them from the agent's identity. The complete chronological prompt record is retained in [ai-use-history.md](#).

Eight selected key prompts and observed outcomes

1. “เริ่มอันนี้เลยอันนี้ LAB4” with the LAB4 PDF: inspected the brief and existing Lab 3 baseline; separated the engineering contract from implementation.
2. “ผมเปลี่ยนโมเดลนะ”: continued the same work; the model change is why exact model names must be confirmed rather than guessed.
3. “จัดการเลย” after discussing the first Issue: created #38/#39 for the specification, UI/API contract and test plan before feature implementation. Peer review required explicit recovery and Start-control coverage.
4. “ส่งไปบอกยัง”: posted the actual contract correction reply and requested Datakung's re-review. Author checks were kept separate from independent approval.
5. “ทำต่อเลย” after foundation approval: implemented staged Actions API/UI with server identity, active assignee, immutable revisions, optimistic concurrency and safe creation recovery. Real browser tests reproduced a committed response loss and a definitively rejected creation response loss.
6. “ต่อเลย” after Actions approval: implemented the atomic resolution gate and recoverable Staff UI. A concurrent unfinished Action can invalidate displayed readiness, so the backend transaction rechecks the condition.
7. “เพื่ออรรีวละ”: read the dashboard changes-requested review, fixed fixture client resolution through server/package.json and passed both requested commands without NODE_PATH. Datakung approved exact head 884df39 and merged #47 into staging.
8. “งั้นทำเลย” after dashboard approval: opened #48 from approved merge 72c5553, added a real 5,000-Ticket/10,000-Action smoke check, ran complete regression/builds and prepared responsive evidence and this nine-part review report. Actual results and remaining release gates are in [final-verification.md](#) and [release.md](#).

My Reflection

AI-assisted draft for the student to read, revise and confirm before submission. It is not a verified statement of the student's personal learning.

การแยกงานกำหนดสเปกออกจากงานเขียนโค้ดช่วยให้เห็นกฎที่โงทียังไม่ได้กำหนดชัด เช่น สถานะของ Action เงื่อนไข Resolve และช่วงเวลา Dashboard ใช้คำนวณ เมื่อระบุ FR, BR และ AC ก่อน จึงสามารถใช้ข้อกำหนดเดียวกันตรวจสอบ API, หน้าจอ และชุดทดสอบได้

ตัวอย่างที่ควรอธิบายให้ได้คือ Ticket Owner กับผู้รับผิดชอบ Action เป็นคนละบทบาท และการซ่อนปุ่ม Resolve อย่างเดียวไม่เพียงพอ เพราะผู้ใช้ส่งคำขอ API โดยตรงหรือมีคนเพิ่ม Action พร้อมกันได้ จึงต้องตรวจเงื่อนไขใน transaction ที่ล็อก Ticket ร่วมกับการแก้ Action

การรีวิวจากเพื่อนทำให้พบข้อผิดพลาดที่การทดสอบบนเครื่องผู้เขียนมองไม่เห็น เช่น Prisma client ของชุดทดสอบ Dashboard ถูกเรียกจากตำแหน่งที่ยังไม่ได้ generate การแก้ไข client ของ server และรันคำสั่งปกติใหม่แสดงว่าหลักฐานต้องทำซ้ำได้ ความสำเร็จบนเครื่องหนึ่งยังไม่เพียงพอ ผลทดสอบบน feature branch ก็ต้องแยกจากผลบน main หลังรีวิวและ merge จริง

ก่อนส่ง ฉันต้องทดลองสาธิตด้วยตนเอง อธิบายเงื่อนไขธุรกิจ ตรวจสอบโมเดลที่ใช้จริง และปรับข้อความสะท้อนคิดนี้ให้ตรงกับสิ่งที่ฉันได้เรียนรู้

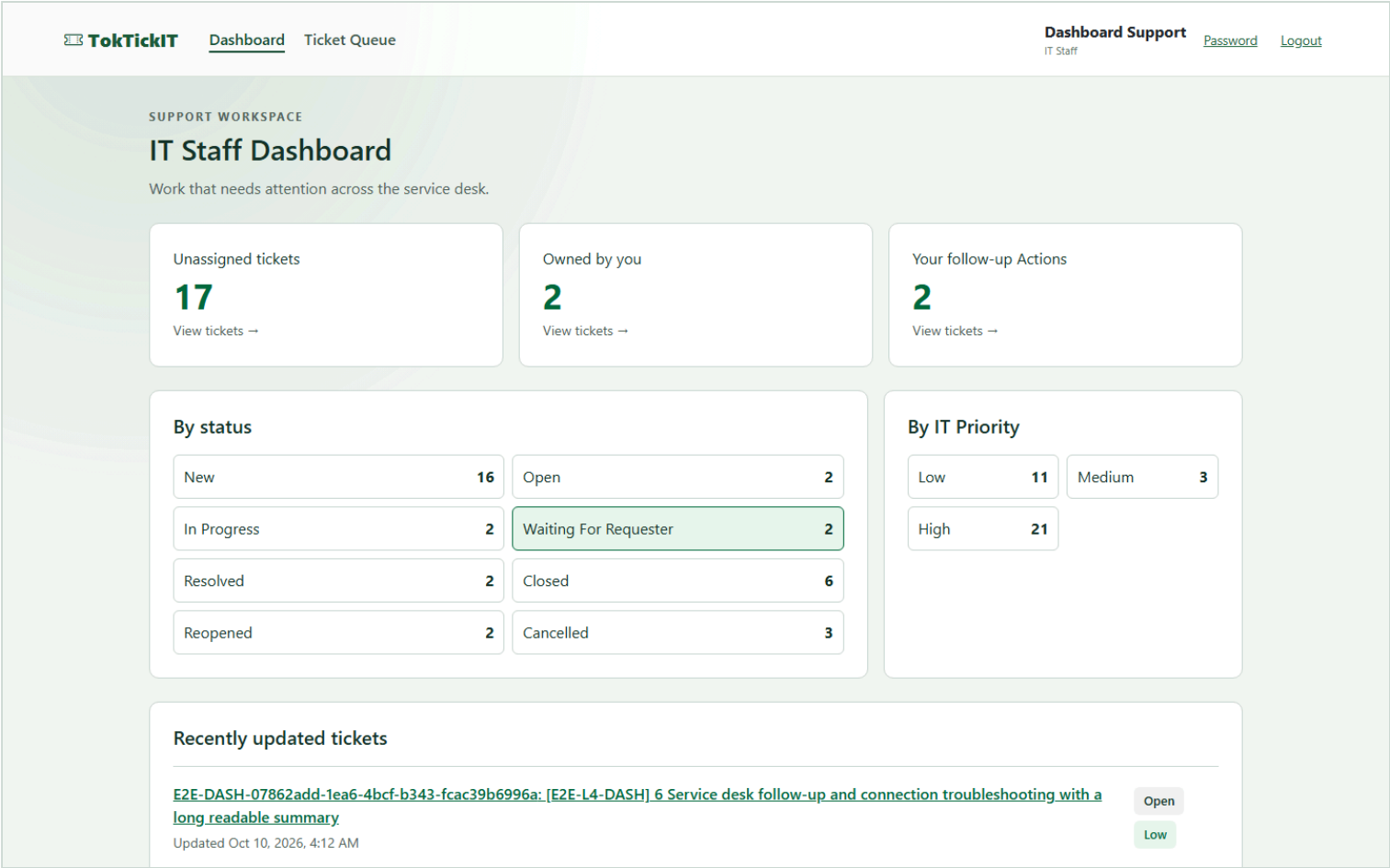
TOKTICKIT / LAB 04 / REVIEW DRAFT

Answer Part 5

Working IT Staff Dashboard UI

Three operational cards, all status/IT Priority counts and at most five recent Tickets are computed by the backend in one repeatable-read snapshot. Follow-ups count assigned unfinished Actions on open Tickets; the drill-down contains distinct Tickets. API/browser tests compare metrics and query intersections. Admin retains Users and its own follow-up scope.

Staff dashboard - desktop



Authoritative counts and concise operational links - top excerpt. [Full-size capture](#)

Captured 2026-10-09T21:13:10.384Z; 5172f83b5470. Synthetic fixture accounts/data. Files labelled simulated use controlled browser responses.

Staff/Admin - tablet and mobile

TokTickIT

Dashboard

Ticket Queue

Dashboard Support

IT Staff

Password

Logout

SUPPORT WORKSPACE

IT Staff Dashboard

Work that needs attention across the service desk.

Unassigned tickets

17

View tickets →

Owned by you

2

View tickets →

Your follow-up Actions

2

View tickets →

By status

New16

Open2

In Progress2

Waiting For Requester2

Resolved2

Closed6

Reopened2

Cancelled3

By IT Priority

Low11

Medium3

High21

Staff tablet - top excerpt. [Full-size capture](#)

TokTickIT

Menu

Dashboard Admin

Administrator

Password

Logout

SUPPORT WORKSPACE

IT Staff Dashboard

Work that needs attention across the service desk.

Unassigned tickets

17

View tickets →

Owned by you

0

View tickets →

Your follow-up Actions

1

View tickets →

By status

New16

Open2

In Progress2

Waiting For Requester2

Resolved2

Closed6

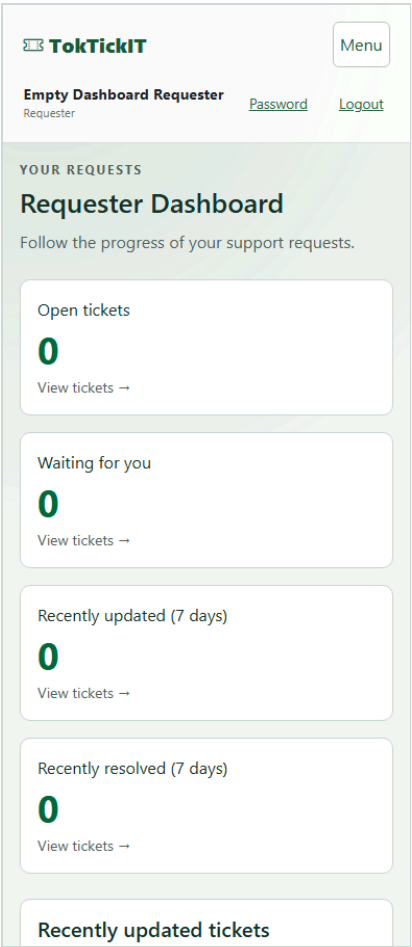
Reopened2

Cancelled3

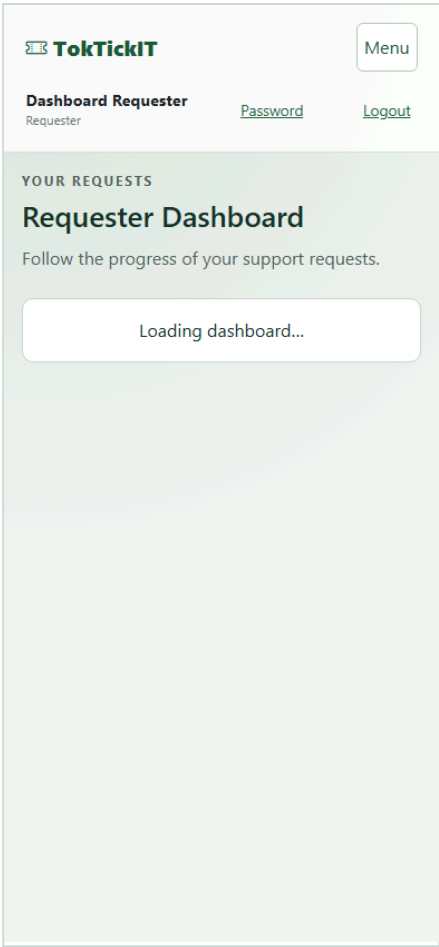
Admin mobile: same operational dashboard - top excerpt. [Full-size capture](#)

Captured 2026-10-09T21:13:10.384Z; 5172f83b5470. Synthetic fixture accounts/data. Files labelled simulated use controlled browser responses.

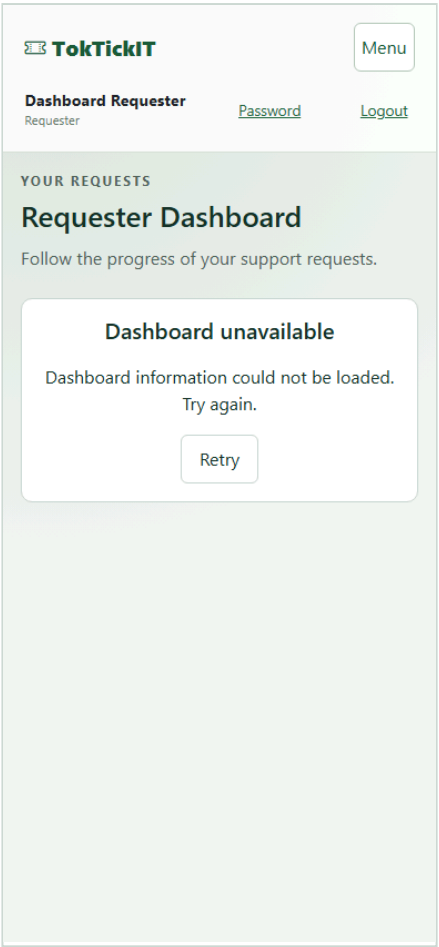
Dashboard feedback: real empty data and simulated response states



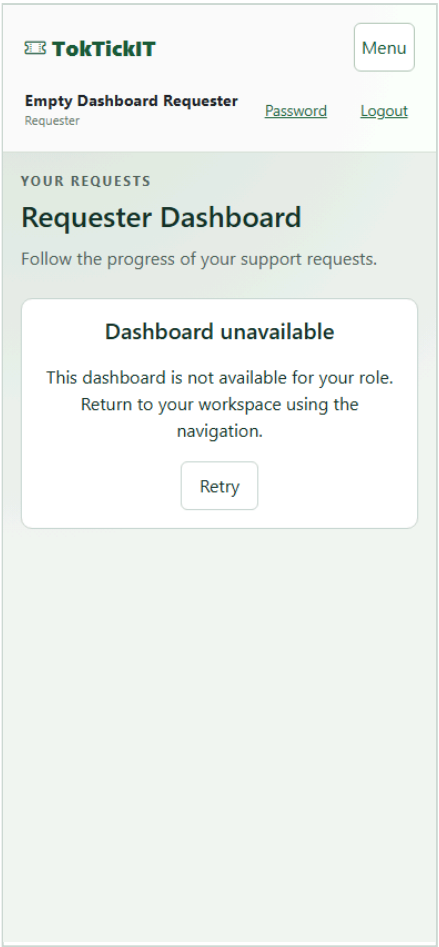
Real empty Requester account - top excerpt. [Full-size capture](#)



Delayed response: loading. [Full-size capture](#)



Simulated 503: safe retry. [Full-size capture](#)



Simulated 403: role feedback. [Full-size capture](#)

Captured 2026-10-09T21:13:10.384Z; 5172f83b5470. Synthetic fixture accounts/data. Files labelled simulated use controlled browser responses.

Answer Part 6

Working Actions Taken UI

Two Staff and Admin record different Actions under one Ticket. Assignee and authenticated performer are distinct. Real APIs enforce active assignees, validation, roles, stale versions, idempotency and atomic append-only revisions. Start/busy/keyboard, completion, cancel and both response-loss recovery paths pass. Requesters see shared fields without mutation/history controls.

Create Action - tablet form

Actions Taken

Work recorded under this ticket. The Ticket Owner coordinates the overall case.

Create Action

Action date and time

10/10/2026 04:12 AM

Action Description

Inspect the connection and retain this draft

Result

Assigned to

Alex Morgan

☒ Follow-Up Required?

Follow-up Note

Ask Requester to verify the restored connection

Attachment Notes

See the diagnostic file attached to this Ticket

Date, description, Result, assignee, follow-up and Attachment Notes - bottom excerpt. [Full-size capture](#)

Captured 2026-10-09T21:13:10.384Z; 5172f83b5470. Synthetic fixture accounts/data. Files labelled simulated use controlled browser responses.

Edit and required-field validation - mobile

Actions Taken

Work recorded under this ticket. The Ticket Owner coordinates the overall case.

Edit Action

Action date and time

10/10/2026 04:12 AM

Action Description

Inspect the connection and retain this draft

Result

Connection checked; follow-up verification remains

Assigned to

Alex Morgan

☒ Follow-Up Required?

Follow-up Note

Ask Requester to verify the restored connection

Attachment Notes

See the diagnostic file attached to this Ticket

Save Action Cancel

Edit preserves values; CREATED and EDITED verified - bottom excerpt. [Full-size capture](#)

Actions Taken

Work recorded under this ticket. The Ticket Owner coordinates the overall case.

Create Action

Action date and time

10/10/2026 04:12 AM

Action Description

Inspect the connection and retain this draft

Result

Assigned to

Alex Morgan

☒ Follow-Up Required?

Follow-up Note

Attachment Notes

See the diagnostic file attached to this Ticket

Save Action Cancel

Missing required follow-up note focuses the field and retains draft - bottom excerpt. [Full-size capture](#)

Captured 2026-10-09T21:13:10.384Z; 5172f83b5470. Synthetic fixture accounts/data. Files labelled simulated use controlled browser responses.

Multiple Actions and Requester read-only visibility

Public Comments

Internal Notes

Visible to the requester

No public comments yet.

Previous

Page 1 of 1

Next

Actions Taken

Add Action

Work recorded under this ticket. The Ticket Owner coordinates the overall case.

Oct 10, 2026, 4:12 AM

Investigated the failing client

CompletedAssigned to Alex Morgan · Performed by Alex Morgan

Result: Reproduced the issue

View Action

Oct 10, 2026, 4:12 AM

Reviewed the documented result

PlannedAssigned to Alex Morgan · Performed by Taylor Admin

View Action

Oct 10, 2026, 4:12 AM

Verified the client settings with another staff member

PlannedAssigned to Alex Morgan · Performed by Jamie Support

Result: Confirmed the connection settings

View Action

Page 1 of 1

Previous

Next

1–2,000 characters · Shared with everyone on this ticket

Post public comment

Actions Taken

Work recorded under this ticket. The Ticket Owner coordinates the overall case.

All Actions

Verify the VPN client

Status

In Progress

Action date

Oct 10, 2026, 4:12 AM

Assigned to

Alex Morgan

Performed by

Alex Morgan

Result

Client configuration checked

Follow-up

Not required

Attachment Notes

None

Two Staff and Admin: multiple retained Actions - bottom excerpt. [Full-size capture](#)

Requester Action detail: mutation controls absent - bottom excerpt. [Full-size capture](#)

Captured 2026-10-09T21:13:10.384Z; 5172f83b5470. Synthetic fixture accounts/data. Files labelled simulated use controlled browser responses.

Start and real competing-version recovery - desktop

Work recorded under this ticket. The Ticket Owner coordinates the overall case.

Action started.

← All Actions

Edit

Complete

Cancel Action

Verify the VPN client

Status	Action date
In Progress	Oct 10, 2026, 4:12 AM
Assigned to	Performed by
Alex Morgan	Alex Morgan
Result	Follow-up
Client configuration checked	Not required
Attachment Notes	
None	
Change history	
1. Created by Alex Morgan · Oct 10, 2026, 4:12 AM	
2. Started by Alex Morgan · Oct 10, 2026, 4:12 AM	

Started state with actor/revision - focused excerpt. [Full-size capture](#)

Captured 2026-10-09T21:13:10.384Z; 5172f83b5470. Synthetic fixture accounts/data. Files labelled simulated use controlled browser responses.

This Action changed. Reload the latest details before trying again.

Reload latest

← All Actions

Edit

Start

Complete

Cancel Action

Verify the VPN client

Status	Action date
Planned	Oct 10, 2026, 4:12 AM
Assigned to	Performed by
Alex Morgan	Alex Morgan
Result	Follow-up
Client configuration checked	Not required
Attachment Notes	
None	
Change history	
1. Created by Alex Morgan · Oct 10, 2026, 4:12 AM	

Real stale version; Reload latest - focused excerpt. [Full-size capture](#)

Answer Part 7

Working Ticket Workflow

All 64 Ticket status edges are verified. Formal Resolve requires at least one COMPLETED Action with a nonblank Result and no unfinished Actions, checked under the same Ticket lock as Action writes. Requester indication is advisory. Admin's Action privileges do not grant Ticket status permission. Closing/reopening retains Action history and stable ordering.

Resolution ready - desktop

Manage ticket

Owner

Alex Morgan

Assign to

Alex Morgan · It Staff

▼

Assign / Reassign

IT Priority

High

▼

Requested priority stays unchanged.

Save priority

Next status

Choose next status

▼

Actions are ready for resolution.

Retry readiness

Update status

Server readiness and Staff status controls -
focused excerpt. [Full-size capture](#)

Captured 2026-10-09T21:13:10.384Z; 5172f83b5470. Synthetic fixture accounts/data. Files labelled simulated use controlled browser responses.

Blocked and competing-write feedback - tablet/mobile

Deterministic requester journey evidence for the Lab 2 quality gate.

Manage ticket

Owner

Alex Morgan

Ticket updated.

Assign to

Alex Morgan · It Staff

Assign / Reassign

IT Priority

High

Requested priority stays unchanged.

Save priority

Next status

Choose next status

Resolution requires a completed Action with a Result and no unfinished Actions. Completed with Result: 0; unfinished: 0.

Retry readiness

Update status

Ticket updated.

Manage ticket

Owner

Alex Morgan

Ticket updated.

Assign to

Alex Morgan · It Staff

Assign / Reassign

IT Priority

High

Requested priority stays unchanged.

Save priority

Next status

Resolved

Resolution requires a completed Action with a Result and no unfinished Actions. Completed with Result: 1; unfinished: 1.

Retry readiness

Update status

Resolve requires at least one completed Action with a Result and no planned or in-progress Actions.

Attachments

Blocked: no documented completed work - focused excerpt. [Full-size capture](#)

A real competing unfinished Action invalidates readiness - focused excerpt. [Full-size capture](#)

Captured 2026-10-09T21:13:10.384Z; 5172f83b5470. Synthetic fixture accounts/data. Files labelled simulated use controlled browser responses.

[Closed-state capture](#); the complete real-browser journey also checks reopening and preserved Action history.

TOKTICKIT / LAB 04 / REVIEW DRAFT

Answer Part 8

Working Requester Dashboard and Final Regression UI

Requester metrics/recent rows are strictly owned; open/waiting cards and recent details lead to real matching screens. Tests verify direct other-user denial, inclusive UTC boundaries, deterministic ordering and one concurrent snapshot. Full retained browser cases cover authentication, My Tickets, files, public/private conversations, Staff and account management.

Requester dashboard - desktop



Dashboard

My Tickets

Create Ticket

Dashboard Requester

Requester

[Password](#)

[Logout](#)

YOUR REQUESTS

Requester Dashboard

Follow the progress of your support requests.

Open tickets

4

View tickets →

Waiting for you

1

View tickets →

Recently updated (7 days)

7

View tickets →

Recently resolved (7 days)

1

View tickets →

Recently updated tickets

[E2E-DASH-07862add-1ea6-4bcf-b343-fcac39b6996a: \[E2E-L4-DASH\] 6 Service desk follow-up and connection troubleshooting with a long readable summary](#)

Updated Oct 10, 2026, 4:12 AM

Open

[E2E-DASH-254067f0-27c0-4fec-87db-bb8decdb6663: \[E2E-L4-DASH\] 5 Service desk follow-up and connection troubleshooting with a long readable summary](#)

Updated Oct 10, 2026, 4:12 AM

In Progress

[E2E-DASH-007a8f05-2d47-4f3f-bfea-2d2f1081e1a5: \[E2E-L4-DASH\] 4 Service desk follow-up and connection troubleshooting with a long readable summary](#)

Updated Oct 10, 2026, 4:12 AM

Cancelled

[E2E-DASH-bb88c8e6-19ce-47aa-b97c-c5fe1245abdf: \[E2E-L4-DASH\] 3 Service desk follow-up and connection troubleshooting with a long readable summary](#)

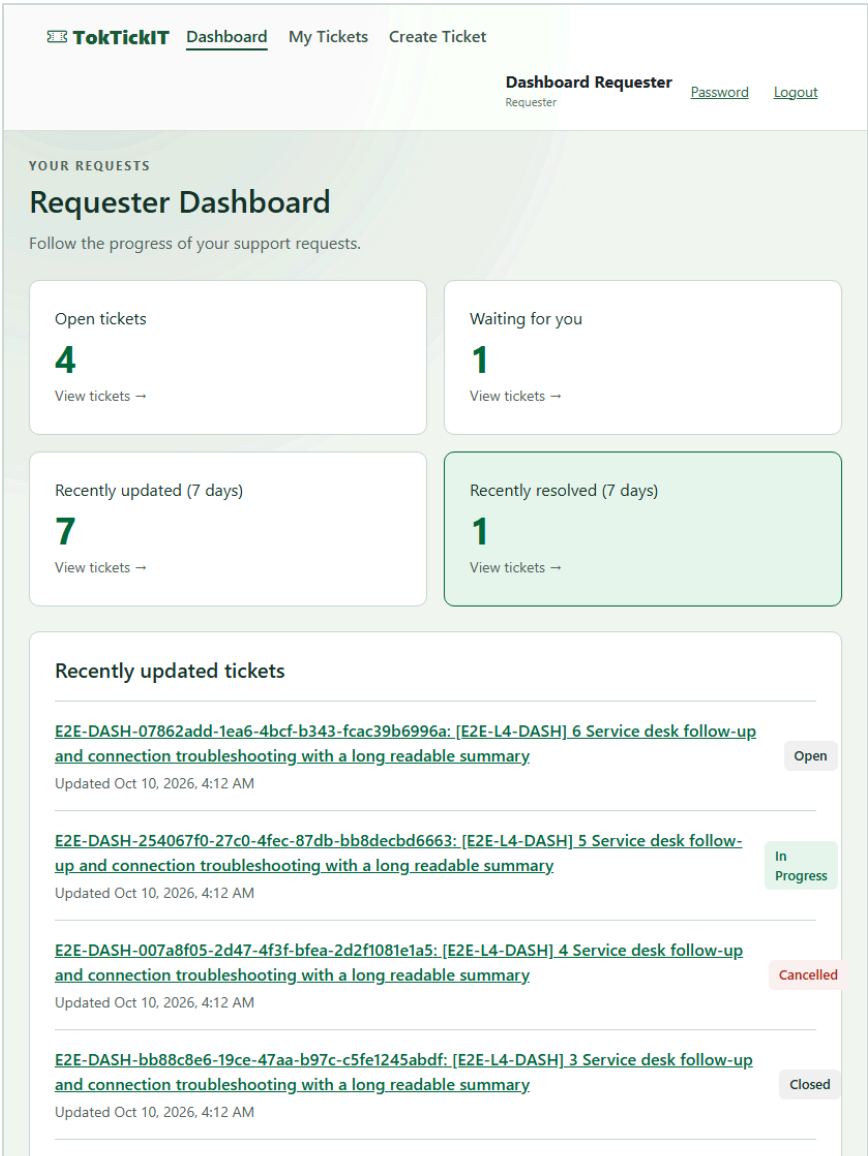
Updated Oct 10, 2026, 4:12 AM

Closed

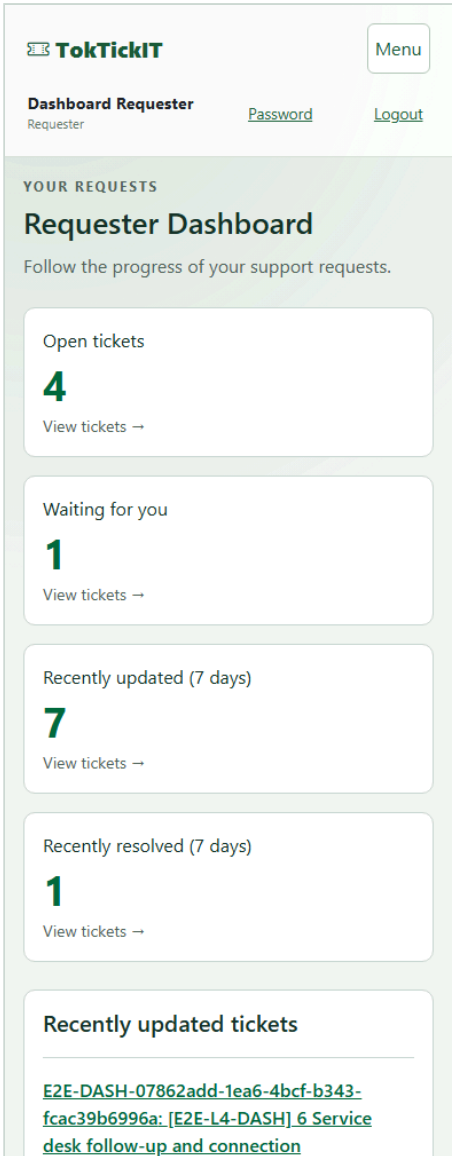
Owned counts and recent/attention-required Tickets - top excerpt. [Full-size capture](#)

Captured 2026-10-09T21:13:10.384Z; 5172f83b5470. Synthetic fixture accounts/data. Files labelled simulated use controlled browser responses.

Requester dashboard - tablet/mobile



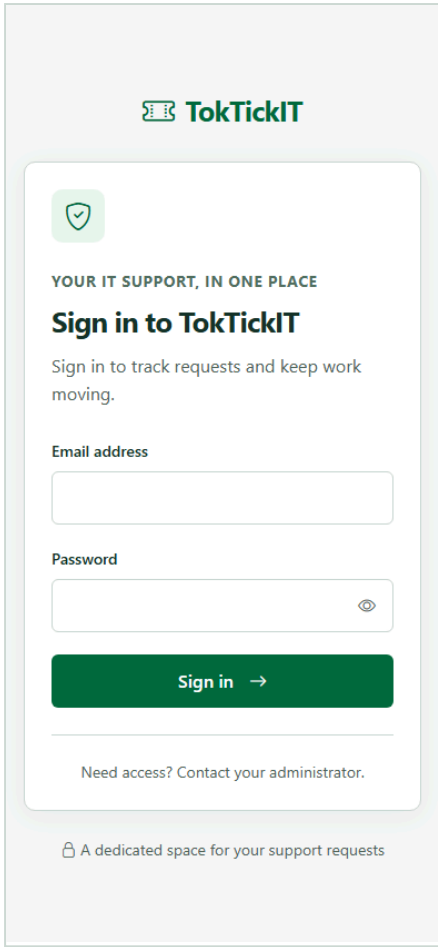
Tablet card wrapping and keyboard focus - top excerpt. [Full-size capture](#)



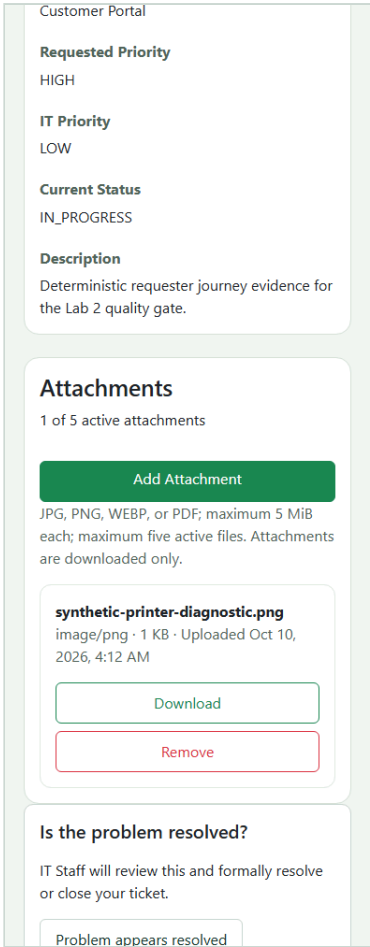
Mobile stacked cards and owned recent rows - top excerpt. [Full-size capture](#)

Captured 2026-10-09T21:13:10.384Z; 5172f83b5470. Synthetic fixture accounts/data. Files labelled simulated use controlled browser responses.

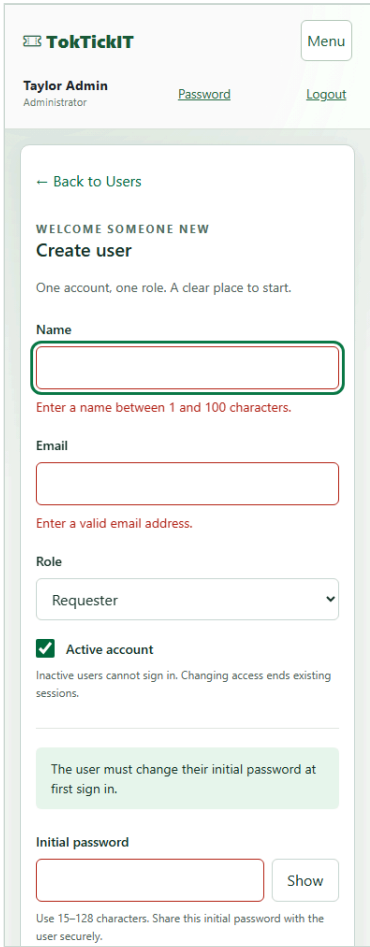
Representative retained regression - mobile



Real login screen. [Full-size capture](#)



Requester attachment/public conversation - focused excerpt. [Full-size capture](#)



Administrator validation placement - top excerpt. [Full-size capture](#)

Captured 2026-10-09T21:13:10.384Z; 5172f83b5470. Synthetic fixture accounts/data. Files labelled simulated use controlled browser responses.

[Private Staff note capture](#) and [real Staff conflict capture](#). The browser output includes all retained responsive cases. Final-main rerun remains pending.

TOKTICKIT / LAB 04 / REVIEW DRAFT

Answer Part 9

Zen Green UI, Responsive, Accessibility and Final Polish

Rendered repository source: <docs/lab-04/ui-spec.md>

Lab 4 UI Specification

Build on the Lab 3 Zen Green tokens and Ticket components. Dashboard navigation is visible only to its role and has `aria-current="page"` on the active route. Server permissions remain authoritative.

Requester/Staff normal sign-in landing is the corresponding dashboard; Admin retains Users. Explicit role-allowed list/detail destinations and their dashboard filters survive authentication. Lists expose Ticket scope and follow-up controls, and clearing filters releases dashboard scopes. Recent/resolved cards identify the seven-day metric window; their destination lists retain the contract's broader sorted/status scope. Multiple qualifying Actions may lead to one Ticket in the follow-up list, with this distinction stated on screen.

Requester dashboard `/dashboard`

Four compact cards: Open Tickets, Waiting for You, Recently Updated (7 days), Recently Resolved (7 days). Below, up to five recently updated owned Tickets show number, summary, status text and update time. Each card has an accessible link to My Tickets with its filter; each row links to owned Ticket Detail. Use zero values and a clear "No recent tickets" message. Show skeleton/status while loading, Retry for safe failure, and Sign in/Return if session or role is invalid. On mobile, cards stack and links remain touch sized.

Staff/Admin dashboard `/staff/dashboard`

Cards: Unassigned, Owned by You, Follow-up Actions; compact status and IT Priority counts; up to five recently updated Tickets. Count links open Ticket Queue or filtered list; recent rows open Staff Ticket Detail. Admin sees the same operational summary and retains User Management navigation. Do not repeat the full queue or expose Internal Notes. Distinguish urgent HIGH priority in text and badge. Empty, loading, forbidden and safe failure states match Requester dashboard.

Actions Taken on Ticket Detail

Show a section after Ticket information/conversation. It lists Action date/time, description, result, status, assignee, performer, follow-up and Attachment Notes. Stable date/ID ordering is visible. Requester mode is read-only. Staff/Admin have Create, View/Edit, Start and Complete/Cancel controls when permitted by the Action/Ticket state. Form fields: date/time, description, result, active-assignee select, Follow-up Required checkbox, conditional Follow-up Note, Attachment Notes. Show field errors next to controls; preserve values after recoverable failures. Save buttons are disabled while a mutation is pending. A stale version displays a conflict banner and Reload action. Completion asks for Result; cancel requires confirmation. History is shown to Staff/Admin as a read-only timeline with actor and time.

Start an Action

Show an explicitly labelled **Start** button in the Action detail controls for Staff/Admin when the Action is PLANNED and its Ticket is neither CLOSED nor CANCELLED. It sends `PATCH /api/tickets/:ticketId/actions/:actionId/status` with `status: IN_PROGRESS` and the displayed `expectedVersion`. The server records the authenticated performer and a STARTED revision. Requesters never receive Start or other mutation controls. IN_PROGRESS, COMPLETED and CANCELLED Actions have no Start control; terminal Actions and CLOSED/CANCELLED Tickets have no Action mutation controls. Backend role/state/version checks enforce the same rules for direct requests.

While Start is pending, show **Starting...**, disable mutation controls and prevent a second click. On success, refresh the selected Action, list/status badge and history from the server, announce **Action started**, and keep keyboard focus usable. On a stale-version 409, retain displayed/entered information, show an announced conflict and offer **Reload latest** before the user retries against the latest version. Other failures show a safe error; retain information and reload current state before retrying an ambiguous request. Planned coverage is UI-05 and E2E-04 in [tests.md](#), including Staff/Admin success, Requester/terminal restrictions, pending protection and conflict recovery.

The Ticket status UI must show permitted transitions. RESOLVED is unavailable until the server gate passes; a server rejection explains that unfinished Actions remain. After a successful mutation refresh Ticket summary and dashboard-relevant counts on navigation. Requester resolution indication remains separate and advisory.

Staff Ticket detail shows server-provided resolution counts/readiness and a Retry readiness button for statuses that can enter RESOLVED. Missing/failed readiness disables RESOLVED; Action changes refresh readiness while retaining owner/priority drafts. A competing unfinished Action may invalidate displayed readiness: ACTIONS_INCOMPLETE refreshes the gate, announces/focuses the error, retains the attempted status and permits explicit readiness retry after the Actions are completed/cancelled. Stale version or an ambiguous status response requires Reload latest details. Status writes have busy/ref protection against duplicate confirmation/submission. Admin Action permissions retain no Ticket status controls.

Responsive and accessibility checks

Use existing Zen Green type/color/spacing and status badges; status meaning includes text. Desktop has compact cards and a table/list, tablet wraps cards, mobile stacks cards and Actions without horizontal page scroll. Form labels are explicit; conditional follow-up note is announced; focus moves to error summary after failed submit and returns to trigger after dialog close. Keyboard users can reach all actions, see focus, and operate dialogs. Loading/success/error updates use appropriate live regions. No clipped long descriptions, overlap or inaccessible modal controls at 1440×900, 834×1112 and 390×844.

Visual review checklist: active navigation; card labels/values; zero/empty states; Action read-only/edit distinction; private/shared content distinction; inline validation; focus visibility; modal focus; badge text; no clipping/overlap/horizontal overflow at three widths. Current integrated captures are under [artifacts/lab-04/release/screenshots/](#); the manifest maps their exact paths and provenance. See the [author inspection checklist](#). Native HTML required-field validation focuses the invalid field; server feedback uses the labelled live/error region. Final-main and student zoom checks remain release gates.

Rendered repository source: [docs/lab-04/visual-accessibility.md](#)

LAB4 Visual and Accessibility Inspection

Author review, 2026-10-10, on feature-branch captures from 5172f83. [Capture manifest](#) records provenance and full-size files. This checklist complements passing component/browser checks; it does not claim complete WCAG certification or the student's personal demonstration.

Check	Evidence and observed scope	Author status
Zen Green shell, active navigation, text/status badges	Requester/Staff/Admin dashboards and retained screens use shared colors, typography, spacing and text cues	Author visual pass: all 75 captures inventoried; representative full-size views checked
Desktop/tablet/mobile	All major LAB4 dashboards, Actions list/detail/forms and resolution feedback captured at 1440/834/390px	Author visual pass at all three widths; no clipping/overlap found in inspected views
Narrow screen and horizontal overflow	Retained Requester/Staff/Admin/system checks include 320px; all new form/dashboard captures assert document width	Browser pass
Keyboard links and visible focus	Real dashboard link activation, filter navigation, Start focus/busy handling and ten real role routes	Browser pass
Native required validation and draft retention	Empty follow-up note focuses the required textarea, reports valueMissing and preserves description; create/edit revision pair confirmed	Browser pass; form capture available
Labels, errors and non-color information	Explicit Action/date/assignee/conditional-note labels, text status and field aria-invalid/aria-describedby; component and style checks retained	Automated and author visual pass

Check	Evidence and observed scope	Author status
Read-only and private/shared distinctions	Requester Action controls absent, Admin Ticket status controls absent; private notes separate and denied to Requester	API/browser pass; captures available
Dialog keyboard operation	Retained Admin dialog focus trap/Escape/return-focus; Ticket/Action confirmation flows retained	Browser pass
Loading, empty, forbidden and safe failure	Dashboard states explicitly captured; browser-simulated loading/503/403 names are labelled; empty account is real owned data	Browser and author visual pass
Conflict and safe recovery	Real competing Action/resolve changes, version reload, response-loss create recovery and retained draft paths	API/component/browser pass
Browser errors and API read failures	Authenticated-route polish test visits ten permitted role routes, asserting no console errors/page errors or failing API reads	Browser pass; intentional pre-login 401 probe excluded
Clipping, overlap and readable fields	Inspect full-size representative desktop/tablet/mobile forms, dashboard and workflow captures, then all PDF pages	Author pass: final report pages rendered/inspected; focused excerpts labelled and full-size links retained
Student native zoom and final demonstration	Student must try normal browser zoom, read Reflection and demonstrate business rules	Pending student confirmation before submission

Simulated Dashboard failure/forbidden captures prove UI feedback only. Separate real API tests establish role denial, safe failure redaction and ownership. Complete image hashes prove file integrity, not visual quality; the latter requires the inspection recorded above.

Visual correction: inspection found a blue, oversized native Follow-up checkbox. Commit 5172f83 scopes its 20px intrinsic size/green accent and keeps the label target 44px. Full regression/build/browser capture were repeated; final mobile create/edit fields are aligned and the label stays on one line. PDF layout iterations removed split figures/captions and selected focused workflow excerpts so readiness and rejection text are visible.

[All 75 captures and checksums](#); [final release checklist](#). The student must confirm model names, revise Reflection, demonstrate the application and review native zoom before the final report is submitted.